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SEMI E116.1-0623 (Reapproved 0324) SPECIFICATION FOR SECS-II PROTOCOL FOR EQUIPMENT PERFORMANCE TRACKING (EPT)

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1 Purpose

1.1 This Standard maps the services and data of the Equipment Performance Tracking (EPT) Standard to SECS-II streams, functions, and data definitions.

2 Scope

2.1 This Standard applies to all implementations of EPT that use the SECS-II message protocol (SEMI E5). Compliance to this Standard requires compliance to both SEMI E116 and SEMI E5.

NOTICE: SEMI Standards and Safety Guidelines do not purport to address all safety issues associated with their use. It is the responsibility of the users of the Documents to establish appropriate safety and health practices, and determine the applicability of regulatory or other limitations prior to use.

3 Referenced Standards and Documents

3.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E39 — Specification for Object Services: Concepts, Behavior, and Services

SEMI E116 — Specification for Equipment Performance Tracking (EPT)

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

4 SECS-II Attribute Definitions

4.1 Table 1 provides the SECS-II structure definitions for the SEMI E116 EPTTracker Object. This must be compliant with the SEMI E39 (OSS) protocol. The host does not have the ability to create or destroy these objects. However, the host must be able to access all attributes via the default SEMI E39 GetAttr services (for all attributes) and SetAtt services (for RW attributes).

4.2 A subset of these object attributes are available via SECS-II Data Variables that may be provided via event-reports. Refer to the section on Variable Mapping in this Standard.

4.3 A subset of these object attributes are available via SECS-II Status Variables. Refer to the section on Variable Mapping in this Standard.

Table 1 EPTTracker Object SECS-II Attributes Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
ObjType	20 (A[10]) must always be set to EPTTracker
ObjID	20 (A[1..80]) <OBJECTIDENTIFIER> ObjID is equipment defined. Examples: EQUIPMENT, CHAMBER A, ROBOT ARM. Conforms to the restrictions of ObjID as specified in SEMI E39.1.

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Attribute Name	*Attribute Data Form: SECS-II Structure*
BlockedReason	51 (U1) BlockedReason where the values are enumerated as: 0 = Not Blocked 1 = Unknown 2 = Safety Threshold 3 = Error Condition 4 = Parametric Exception 5 = Aborting, Aborted 6 = Pausing, Paused 7 = Reserved 8 = Reserved 9 = Reserved
BlockedReasonText	20 (A) BlockedReasonText Usage is equipment defined. Examples: Abort – Host-requested Abort, Fault – Reticle Stage Error; Pause – Host-requested Pause
EPTElementType	51 (U1) EPTElementType where the values are enumerated as: 0 = Equipment 1 = Production EPT Module 2 = EFEM/Load Port EPT Module
EPTState	51 (U1) EPTState where the values are enumerated as: 0 = Idle 1 = Busy 2 = Blocked
EPTStateTime	54 (U4) EPTStateTime
EPTElementName	20 (A) EPTElementName Usage is equipment defined. Examples: EQUIPMENT, CHAMBER A, ROBOT ARM.
PreviousEPTState	51 (U1) PreviousEPTState where the values are enumerated as: 0 = Idle 1 = Busy 2 = Blocked 3 = No State
PreviousTaskName	20 (A[0..80]) PreviousTaskName Usage is equipment defined. Examples: Loading, Etching, Pumping Down
PreviousTaskType	51 (U1) PreviousTaskType where the values are enumerated as: 0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
TaskName	20 (A[0..80]) TaskName Usage is equipment defined. Examples: Loading, Etching, Pumping Down

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Attribute Name	*Attribute Data Form: SECS-II Structure*
TaskType	51 (U1) TaskType where the values are enumerated as: 0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
TransitionTimestamp	20 Timestamp format as specified by the TimeFormat equipment constant value setting (refer to SEMI E5).
Transition	51 (U1) Transition where the values are enumerated as: 1 = Transition 1 2 = Transition 2 . . . 9 = Transition 9
TrackerEventID	54 (U4) or 50 (U8) TrackerEventID where: This is identical to EPTStateChangeEventi corresponding to this object (refer to § 5).
DisableEventOnTransition	L,n 1. 51 (U1) <Transition Number> 2. 51 (U1) <Transition Number> … n. 51 (U1) <Transition Number> where the elements of this list are can only have values in the range 1..9, in order, with no repeats. This list may be empty.

5 Collection Event Mapping

5.1 Table 2 shows the specific SECS-II streams and functions that shall be used for SECS-II implementations of the events defined in EPT. SEMI E116 defines two collection event implementations. Both implementations can coexist simultaneously even though two collection events are triggered at the equipment for each EPT object's state change. A host can enable and disable collection events to decide which collection events are reported to the host.

5.1.1 In the first implementation, a single collection event is generated for each of the equipment and the modules to report transitions occurring within the state model. This approach differs from state models that are defined in other related SEMI Standards, which utilize a unique CEID for each transition in the state model. For example, a piece of equipment with 3 EPT modules would have 4 associated CEIDs. The CEID indexed 0 would report equipment-level events and 3 CEIDs corresponding to the equipment's 3 EPT modules indexed 1…3 would report EPT module-level events.

5.1.2 In the second implementation, a unique collection event is defined for each EPT module transition and for each EPT equipment state transition. This approach matches the approach in other GEM 300 related SEMI Standards. For the EPT equipment, a unique collection event is defined for each state transition. For the EPT modules, a unique collection event is defined for each state transition, which is shared by all EPT modules. An equipment always implements the nine EPTEquipmentTransition collection events for the equipment. Regardless of the number of EPT modules, an equipment also always implements the nine EPTModuleTransition collection events, which are shared by all EPT modules. Data variables EPTObjID and EPTElementName allow the host to distinguish between transitions from each EPT module.

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Table 2 Event Mapping Table

Event Name	*Stream, Function*	*SECS-II Message Name*
EPTStateChangeEventi	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition1	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition2	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition3	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition4	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition5	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition6	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition7	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition8	S6F11/F12	Event Report Send/Acknowledge
EPTEquipmentTransition9	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition1	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition2	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition3	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition4	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition5	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition6	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition7	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition8	S6F11/F12	Event Report Send/Acknowledge
EPTModuleTransition9	S6F11/F12	Event Report Send/Acknowledge

6 Variable Mapping

6.1 Table 3 shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E116 EPT data and status variables. The variables must be updated as specified in the SEMI E116 Standard when a transition occurs in the state-model for any of the EPTTracker objects implemented on the equipment. The Variable Names below match the object Attribute Names from which the variables are to be updated (refer to Table 1).

6.1.1 These variables are tied to the occurrence of transitions in the state model and are not impacted by the DisableEventOnTransition attribute in the EPTTracker object.

Table 3 Variable Item Mapping Table

Variable Name	*Class*	*Format*
EPTObjID	DV	20 (A[1..80])
EPTObjID_i	SV	20 (A[1..80])
EPTElementName	DV	20 (A[1..80])
EPTElementName_i	SV	20 (A[1..80])
EPTElementType	DV	51 (U1)
EPTElementType_i	SV	51 (U1)
EPTState	DV	51 (U1)
EPTState_i	SV	51 (U1)
PreviousEPTState	DV	51 (U1)
PreviousEPTState_i	SV	51 (U1)
EPTStateTime	DV	54 (U4)

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Variable Name	*Class*	*Format*
EPTStateTime_i	SV	54 (U4)
TaskName	DV	20 (A[0..80])
TaskName_i	SV	20 (A[0..80])
TaskType	DV	51 (U1)
TaskType_i	SV	51 (U1)
PreviousTaskName	DV	20 (A[0..80])
PreviousTaskName_i	SV	20 (A[0..80])
PreviousTaskType	DV	51 (U1)
PreviousTaskType_i	SV	51 (U1)
BlockedReason	DV	51 (U1)
BlockedReason_i	SV	51 (U1)
BlockedReasonText	DV	20 (A[0..80])
BlockedReasonText_i	SV	20 (A[0..80])
TransitionTimestamp_i	SV	20 (A)
Transition_i	SV	51 (U1)
TrackerEventID_i	SV	54 (U4) or 50 (U8)
NumberOfEPTModules	SV	54 (U4)

Table 4 Additional Data Item Mapping Table

Variable Name	*SECS-II Data Item*
Clock	CLOCK

#1 Refer to SEMI E5 for these variable definitions.

7 Related Documents

7.1 *SEMI Standards and Safety Guidelines*

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

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